

Design a News Feed System

News Feed System

A news feed system is one of the most common and important large-scale system design interview questions.

Similar interview questions include:

- Design Facebook News Feed
- Design Instagram Feed
- Design Twitter Timeline
- Design LinkedIn Feed

What is a News Feed?

According to the Facebook help page:

None

News Feed is the constantly updating list of stories shown on the user's homepage.

The feed contains updates from:

- Friends
- Pages
- Groups
- Followed accounts

Types of Content in News Feed

A news feed may contain:

- Text posts
- Photos
- Videos
- Links
- Reactions/Likes
- Comments
- Shares
- Stories
- App activities

Main Purpose of News Feed

The primary goal is:

None

Show relevant and updated content to users efficiently.

Core Functionalities of a News Feed System

A news feed system mainly supports two operations:

1. Feed Publishing

Users can:

- Create posts
- Upload photos/videos
- Share content

The system distributes content to followers/friends.

2. News Feed Retrieval

Users can:

- Open homepage/feed
- View recent posts
- Scroll continuously

The system fetches personalized feed content.

Key Challenges in Designing News Feed

Designing a scalable news feed system is challenging because:

- Millions of users post simultaneously
- Feed updates happen continuously
- Reads are extremely high
- Feed ranking is complex
- Low latency is required

Important Design Requirements

A good news feed system should support:

Requirement	Description
Scalability	Handle millions/billions of users
Low Latency	Fast feed loading

Reliability	No data loss
High Availability	Feed always accessible
Personalization	Different feed per user
Real-time Updates	New posts appear quickly
Fault Tolerance	System survives failures

Main Components of News Feed System

The high-level architecture generally includes:

- Feed Publisher
- Feed Generator
- Feed Cache
- Database
- Ranking System
- Timeline Storage
- Fanout Service
- API Servers

Two Major Design Approaches

News feed systems commonly use:

1. Fanout on Write (Push Model)

When a user posts:

- The post is pushed to followers immediately

Advantages

- Fast feed reads
- Low read latency

Disadvantages

- Expensive for celebrities with millions of followers

2. Fanout on Read (Pull Model)

Feed is generated when users request it.

Advantages

- Saves write cost
- Better for celebrity accounts

Disadvantages

- Slower reads

Hybrid Approach

Modern systems usually combine both approaches.

Example:

- Normal users → Push model
- Celebrities → Pull model

This improves scalability and efficiency.

Common System Design Topics Covered

This chapter typically discusses:

- Feed publishing flow
- Feed retrieval flow
- Cache design
- Feed ranking
- Data model
- Fanout strategies
- Scalability
- Storage optimization
- Celebrity problem
- Feed consistency

Real-World Examples

Popular applications using news feed systems:

- Facebook
- Instagram
- Twitter/X
- LinkedIn
- Reddit

Key Takeaway

A news feed system is essentially:

None

A highly scalable distributed system that delivers personalized content efficiently and in real time.